A Connected Leakage-Free Hybrid Quantum-Classical Pipeline for Bank Term Deposit Prediction, Constrained Cohort Selection, and Quantum Outcome Simulation

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Quantum Computing & Machine Learning Track · Atomiq QFF 2026 IEM Hackathon

Repository: https://github.com/pradyotqc/Atomiq_QFF2026_IEM · Reproducible Stack: Python 3.12.11 & Qiskit 2.6.2 (seed = 42)

***Abstract*—We present a leakage-free three-stage hybrid quantum-classical bank marketing pipeline verified via `Qiskit Aer` simulation. Across 300 training rows (11.67% base rate, `duration` purged), zero-centering a 6-qubit `Z+ZZ` Pauli kernel with fold-isolated spectral whitening (`Quantum_Kernel_LogReg`) eliminates phase concentration, achieving the lowest Brier score (0.0993), ECE $= 0.0043$, and `PR-AUC` $= 0.2406$ alongside top-ranked `Classical_LogReg` (`PR-AUC` $= 0.2453$, `ROC-AUC` $= 0.6527$). Using verbatim Stage 1 probabilities ($K = 6, C = 25, H = 22$), 20-qubit QUBO `CVaR-QAOA` and 10-qubit Dicke `XY-QAOA`/`HW-VQE` recover the exact global optimum `1110001110` (`gap` $= 0$, net value 23.1835, +24.40% uplift; 13/25 budget [52%], 16/22 capacity [73%] used). A depth-2 6-qubit R_y circuit matches exact Poisson-Binomial math ($P(S \geq 3) = 0.282049$) and cuts simulated readout bias by 95.9% (to 0.282232).**

***Index Terms*—Quantum machine learning, quantum fidelity kernels, kernel concentration, QUBO Ising mapping, Dicke-state XY-QAOA, CVaR-VQE, Poisson-Binomial distribution, readout error mitigation, Qiskit.**

I. Introduction

Direct marketing campaigns for bank term deposit subscriptions require jointly optimizing three tightly coupled decisions under uncertainty: (1) accurately estimating calibrated subscription probabilities $p_i \in [0,1]$ from historical demographic and macroeconomic attributes prior to dialing; (2) selecting a constrained cohort of customers that maximizes expected net return rather than raw propensity score; and (3) quantifying the probability that realized subscriptions clear operational campaign targets. When engineered piecemeal classically, Stage 1 classifiers often prioritize area under the ROC curve without calibrating probability magnitudes, Stage 2 outreach defaults to greedy top-K sorting that neglects asymmetric contact costs c_i, expected deposit sizes d_i, and advisor capacity weights w_i, and Stage 3 relies strictly on approximate sampling.

The Atomiq QFF 2026 IEM Hackathon formalizes this industrial workflow as an integrated benchmark on `bank_campaign_participant_release.xlsx`, explicitly grading quantum rigor (35%), cross-stage accuracy and integration (30%, requiring $< 10^{-12}$ verbatim coupling across CSV deliverables), reproducible software engineering (20%), and technical documentation (15%) [1]. Meeting these specifications required overcoming four interlocking technical obstacles:

1. *Target Imbalance and Post-Call Leakage:* Only 35 of 300 labeled training customers subscribed (11.67% base rate), and post-call variables (specifically `duration`) or dataset-wide feature transformations induce fatal data leakage.
2. *Quantum Kernel Concentration:* Passing zero-mean standardized features into Qiskit's default `zz_feature_map` with offset phases $2(\pi - x_i)(\pi - x_j)$ concentrates off-diagonal fidelity Gram entries toward flat random noise (`PR-AUC` $= 0.1391$) [3].
3. *Slack Qubit Explosion in QUBO Formulations:* Inequality constraints on integer budget and advisor capacity append 10 binary slack variables to a 10-customer decision vector, inflating Hilbert space dimension from $2^{10} = 1,024$ to $2^{20} = 1,048,576$ and diluting shallow QAOA sampling mass.
4. *Gate and Readout Noise Distortion:* Realistic superconducting depolarizing gate noise and asymmetric qubit measurement errors distort combinatorial optimum hit rates and bias multi-customer campaign distributions unless explicitly mitigated.

To resolve all four challenges cohesively, we built a deterministic (`seed = 42`) Python package (`src/`) driven by `run_pipeline.py` inside a pinned Python 3.12.11 virtual environment and verified end-to-end by an automated `unittest` suite and executed Jupyter notebook. Conceptually, the workflow operates as a strict three-step funnel: **Stage 1** estimates honest customer subscription odds, **Stage 2** allocates limited outreach dollars and advisor capacity to assemble the most profitable cohort of six leads, and **Stage 3** simulates campaign-wide conversion risk under exact, shot-limited, and noisy regimes. Fig. 1 illustrates this dataflow and exact artifact coupling.

The remainder of this report is organized as follows. Section II details workbook validation, the anti-leakage firewall, and domain feature engineering. Section III presents Stage 1 classical/quantum models, our kernel concentration fix, and cross-validation benchmarks. Section IV derives the 20-qubit QUBO Ising Hamiltonian, slack-free 10-qubit Dicke `XY-QAOA`, and Stage 2 economic uplift. Section V presents Stage 3 6-qubit circuit mapping, Poisson-Binomial equivalence, and tensor readout mitigation. Section VI audits transpiled gate telemetry, pinned seeds, and hackathon requirements compliance. Section VII concludes with limitations, error analysis, and business deployment

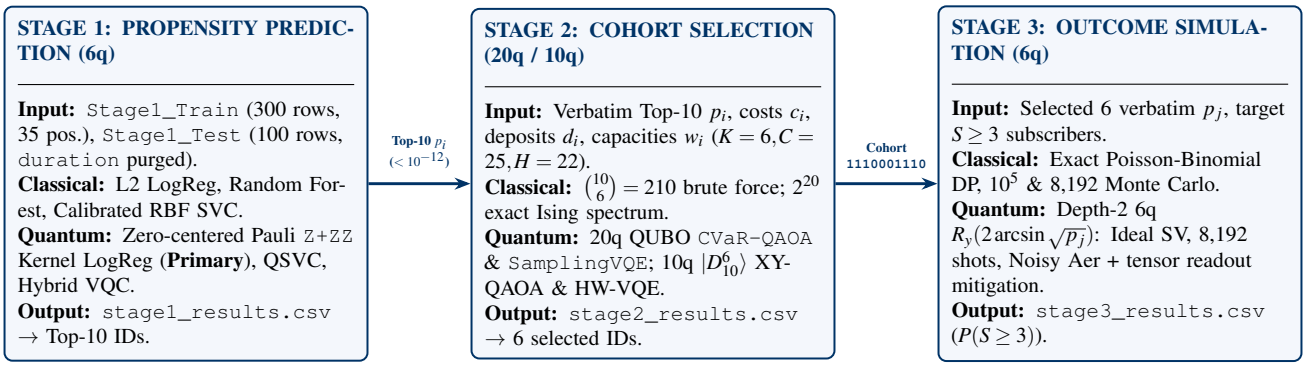


Figure 1. Connected three-stage hybrid quantum-classical architecture. Stage 1 outputs verbatim calibrated probabilities into `stage1_results.csv`; Stage 2 solves exact constrained expected net-value maximization over Top-10 candidates; Stage 3 encodes the selected six customers onto six qubits to estimate $P(\text{subscribers} \geq 3)$ across exact, finite-shot, noisy, and mitigated regimes.

recommendations.

II. Dataset Validation, Anti-Leakage Firewall, and Feature Engineering

A. Workbook Schema and Target Imbalance

All source data reside in `bank_campaign_participant_release.xlsx`, loaded and schema-validated by `load_campaign_data()` in `src/data_loader.py`. The workbook contains five sheets: `README`, `Stage1_Train` (300 labeled records with target `y` $\in$ {`yes`,`no`}), `Stage1_Test` (100 blind customer records), `Stage2_Data` (contact cost c_i, expected deposit d_i, advisor capacity w_i, and `customer_segment` for test IDs), and `Stage2_Constraints` ($K = 6$, budget $C = 25$, capacity $H = 22$). In `Stage1_Train`, 35 customers subscribed (11.67%) and 265 declined (88.33%). Upon ingestion, `load_campaign_data()` asserts exact row cardinalities (300 train, 100 test), zero duplicate `customer_id` entries, and 100% ID coverage inside `Stage2_Data`.

B. Zero-Leakage Firewall Design

Strict operational discipline ensures that no target-leaking or held-out test signal contaminates Stage 1 propensities:

1. *Post-Call Variable Ban:* In Portuguese bank telemarketing benchmarks [2], call `duration` strongly predicts outcome yet is unknown prior to dialing. `data_loader.py` explicitly asserts that `"duration"` is absent from both `train_df` and `test_df`, raising a `ValueError` if present.

2. *Stateless Row-Wise Transforms:* `BankFeatureEngineer` computes purely row-wise nonlinear transformations with zero cross-row statistics.

3. *Fold-Isolated Preprocessing and Tuning:* Across 5-fold Stratified Cross-Validation (`CV_N_SPLITS = 5`, `CV_SPLIT_SEED = 42`), median/modal imputation (`SimpleImputer`), one-hot encoding (`OneHotEncoder(handle_unknown="ignore")`), z-score scaling (`StandardScaler`), univariate ANOVA selection (`SelectKBest(f_classif, k=6)`), kernel eigen-decomposition, and Platt calibration are fit strictly inside each training fold before transforming held-out validation splits.

4. *Single-Shot Test Inference:* Only after all CV benchmarks conclude does the final pipeline refit on full `Stage1_Train` and predict `Stage1_Test` once, sorting descending by `predicted_probability` with deterministic ascending `customer_id` mergesort tie-breaking.

C. Domain Feature Engineering

From raw demographics (`age`, `job`, `marital`, `education`, `default`, `balance`, `housing`, `loan`), campaign contact metadata (`contact`, `day`, `month`, `campaign`, `pdays`, `previous`, `poutcome`), `BankFeatureEngineer` constructs 21 numeric columns. In banking campaigns, extreme account balances span orders of magnitude (including negative overdrafts), uncontacted leads carry a sentinel code pdays $= -1$, and calendar dates wrap around year/month boundaries. Signed logarithmic compression and binary indicator flags keep these signals well-behaved for linear and quantum feature maps:

$$\text{was_prev} = \mathbb{I}(\text{pdays} \geq 0), \quad \text{pdays_clean} = \max(\text{pdays}, 0), \tag{1}$$

$$\text{log1p_pdays} = \ln(1 + \text{pdays_clean}), \tag{2}$$

$$\text{slog_bal} = \text{sign}(\text{balance}) \ln(1 + |\text{balance}|), \tag{3}$$

$$\text{bal_per_age} = \frac{\text{slog_bal}}{\max(\text{age}, 18)}, \tag{4}$$

$$\text{log1p_camp} = \ln(1 + \max(\text{campaign}, 0)), \tag{5}$$

$$\text{log1p_prev} = \ln(1 + \max(\text{previous}, 0)), \tag{6}$$

together with cyclical Fourier calendar encodings to preserve December–January and end-of-month continuity:

$$m_{\sin}, m_{\cos} = \sin\left(\frac{2\pi m_{\text{idx}}}{12}\right), \cos\left(\frac{2\pi m_{\text{idx}}}{12}\right), \tag{7}$$

$$d_{\sin}, d_{\cos} = \sin\left(\frac{2\pi(\text{day} - 1)}{31}\right), \cos\left(\frac{2\pi(\text{day} - 1)}{31}\right), \tag{8}$$

and four domain flags: `has_prev_success`, `has_prev_failure`, `has_known_contact`, and `any_loan`. Within `build_tabular_preprocessor()`, numeric features undergo median imputation and standard

scaling, while all nine categorical fields undergo most-frequent imputation and unknown-safe one-hot encoding.

III. Stage 1: Quantum Kernel Prediction and Calibration

A. Diagnosing and Resolving Quantum Kernel Concentration

Matching a 6-qubit quantum register requires compressing preprocessed features via fold-isolated `SelectKBest(f_classif, k=6)` and standardization ($z_j \sim \mathcal{N}(0,1)$). At first glance, feeding standardized inputs directly into Qiskit's stock `zz_feature_map` appears natural, yet produces near-random results (`PR-AUC` $= 0.1391$, `ROC-AUC` $= 0.5145$). Why does it fail? Qiskit's default map rotates entangling pairs by $2(\pi - x_i)(\pi - x_j)$ [4]; when standardized inputs lie near zero ($x_i \approx 0$), every two-qubit phase sits near $2\pi^2$, scrambling the $2^6 = 64$ computational amplitudes. Consequently, the quantum similarity score between any two distinct customers

$$K_{ij} = \left|\langle \phi(\mathbf{x}_i) \mid \phi(\mathbf{x}_j)\rangle\right|^2 \tag{9}$$

collapses toward flat uniform noise (exponential kernel concentration [3]), erasing discriminative signal.

We cure this geometry collapse in `src/stage1_predict.py` via two coordinated mechanisms:

1. *Smooth Bounded Angle Compression:* `QuantumAngleCompressor` applies smooth hyperbolic tangent compression

$$x_j = b\tanh\left(\frac{z_j}{2}\right) \in (-b, b), \tag{10}$$

with $b = 0.7$ (`Quantum_Kernel_LogReg`, `Hybrid_VQC`) and $b = 0.6$ (`Quantum_Kernel_SVC`), preventing phase wrap-around while maintaining sensitivity near $z_j = 0$.

2. *Zero-Centered Pauli Mapping:* `_build_zero_centered_feature_map` instantiates `pauli_feature_map` with $n = 6$ qubits, reps $= 1$, `paulis=["Z", "ZZ"]`, linear nearest-neighbor entanglement (requiring 0 SWAP gates), $\alpha = 1.0$, and custom `data_map_func` setting $\theta_j = x_j$ on single-qubit Z rotations and $\theta_{jk} = x_j x_k$ on nearest-neighbor ZZ entanglers. Zero-centered angles eliminate artificial π-offset phase scrambling and preserve $K(\mathbf{x}, \mathbf{x}) = 1$.

B. Classifier Portfolio: Three Quantum/Hybrid vs. Three Classical

All six models are evaluated identically inside `run_stage1()` across identical stratified CV splits:

- **`Quantum_Kernel_LogReg` (Primary Model):** Inside each training fold, the positive-semidefinite training Gram matrix K_{train} from Eq. (9) is spectrally decomposed via $K_{\text{train}} = V\Lambda V^\top$. Filtering eigenvalues $\lambda_m > 10^{-8}$, training samples map to explicit orthonormalized RKHS features $F_{\text{train}} = K_{\text{train}} V_{\text{keep}} \Lambda_{\text{keep}}^{-1/2}$, fit with L2-regularized `LogisticRegression(C=2.0)`. Validation and test instances evaluate $F_{\text{eval}} = K_{\text{eval}} V_{\text{keep}} \Lambda_{\text{keep}}^{-1/2}$, producing inherently calibrated maximum-likelihood probabilities.
- **`Quantum_Kernel_SVC`:** Evaluates precomputed quantum fidelity matrices inside `SVC(kernel="precomputed", C=2.0)`, calibrated cleanly via inner 5-fold Stratified OOF Platt scaling ($\sigma(af(\mathbf{x}) + b)$).
- **`Hybrid_VQC`:** Composes the zero-centered 6-qubit Pauli map with a linear 2-layer `RealAmplitudes` variational ansatz (18 trainable rotation parameters) and measures balanced magnetization $\mathcal{O} = \frac{1}{6}\sum_{q=0}^{5} Z_q$. Parameters (θ, w, b) minimize L2-regularized binary cross-entropy ($\lambda_{\text{reg}} = 10^{-2}$, initialized at the empirical log-odds base rate) via derivative-free `COBYLA(maxiter=100)`.
- **Classical Baselines:** L2 `Classical_LogReg(C=0.25)`, `Classical_RF` (300 trees, `max_depth=5`, `min_samples_leaf=5`), and `Classical_RBF_SVC(C=1.0)` calibrated with 5-fold inner `CalibratedClassifierCV`.

C. Stage 1 Comparative Analysis, Reading Fig. 2, and Aer Verification

Table I lists all seven out-of-fold (OOF) and fold-mean metrics computed in `src/metrics.py`, plotted in Fig. 2 and Fig. 3. To interpret Fig. 2 clearly, each of the four bar charts evaluates the same six models side-by-side—slate gray bars denote the three classical baselines, the emerald green bar highlights our selected Primary model (`Quantum_Kernel_LogReg`), and blue bars represent alternative quantum/hybrid candidates:

1. *How to Read the Four Panels of Fig. 2:* **Top-Left (`PR-AUC`, higher is better)** measures precision–recall quality on the imbalanced 11.67% positive cohort; `Quantum_Kernel_LogReg` achieves `0.2406` (2.06× the random baseline `0.1167`), essentially matching linear `Classical_LogReg` (`0.2453`). **Top-Right (`ROC-AUC`, higher is better)** captures global ranking separation; `Classical_LogReg` leads (`0.6527`), while `Quantum_Kernel_LogReg` (`0.6075`) leads all quantum models. **Bottom-Left (`Precision@10`, higher is better)** counts true subscribers among the top-10 ranked customers fed to Stage 2; both margin-based SVMs reach `0.600` (6/10 hits), while `Quantum_Kernel_LogReg` scores `0.400`. **Bottom-Right (`Brier Score`, zoomed $[0.095, 0.105]$, lower is better)** measures mean squared probability error $(p_i - y_i)^2$: here the green `Quantum_Kernel_LogReg` bar sits **lowest overall** at `0.0993`. Meanwhile, `Hybrid_VQC` lags (`ROC-AUC` $= 0.5214$, `PR-AUC` $= 0.1687$) due to slow gradient-free `COBYLA` optimization in 100 steps over 20 variational parameters.
2. *Why `Quantum_Kernel_LogReg` Was Selected as Primary:* Stage 2 computes expected net value ($p_i d_i - c_i$) and Stage 3 computes multi-customer tail probability ($P(S \geq 3)$) directly from the numerical values of p_i. An overconfident ranking model inflates revenue projections and misallocates budget. As Table I and Fig. 3 confirm, while `Classical_LogReg` retains the highest overall ranking (`PR-AUC` $= 0.2453$, `ROC-AUC` $= 0.6527$), `Quantum_Kernel_LogReg` is chosen specifically for probability calibration rather than ranking su-

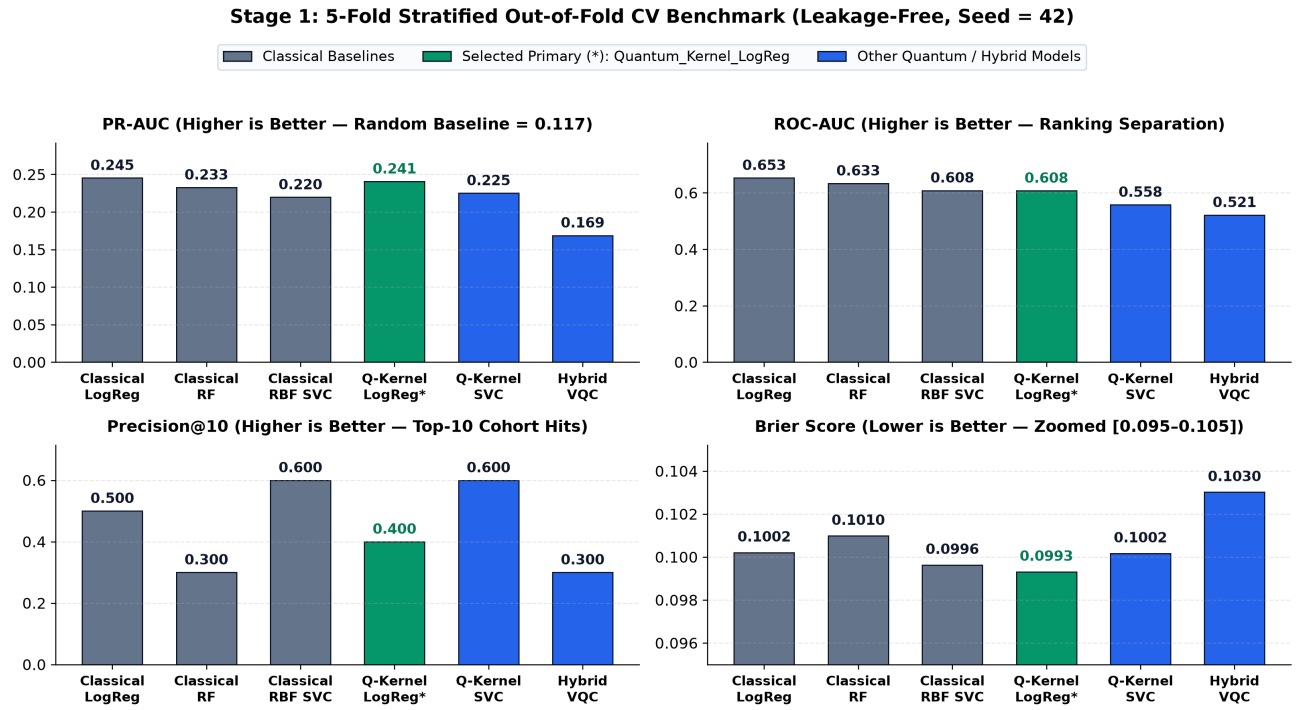


Figure 2. Stage 1 out-of-fold 5-fold Stratified Cross-Validation benchmark comparing three classical baselines (slate gray), the selected primary model `Quantum_Kernel_LogReg` (emerald green), and two alternative quantum/hybrid classifiers (blue) across `PR-AUC`, `ROC-AUC`, `Precision@10`, and zoomed `Brier Score` $[0.095, 0.105]$. Replacing default π-offset phases with zero-centered Pauli encoding lifts `Quantum_Kernel_LogReg` to 2.06× random baseline `PR-AUC` (0.241 vs. 0.117) and lowest Brier score (0.0993).

TABLE I
STAGE 1 OUT-OF-FOLD AND FOLD-WISE 5-FOLD STRATIFIED CROSS-VALIDATION METRICS (`Stage1_Train`, $N = 300$, 35 POSITIVES)

Classifier	ROC-AUC (OOF / Fold ± std)	PR-AUC (OOF / Fold ± std)	Prec@10	Rec@10	Brier Score	Log Loss	ECE (5-bin)
`Classical_LogReg`	**0.6527** (0.6534 ± 0.1258)	**0.2453** (0.3273 ± 0.0583)	0.500	0.1429	0.1002	0.3584	0.0309
`Classical_RF`	0.6332 (0.6334 ± 0.0672)	0.2325 (0.2625 ± 0.0837)	0.300	0.0857	0.1010	0.3500	0.0151
`Classical_RBF_SVC`	0.6083 (0.6237 ± 0.0595)	0.2195 (0.3006 ± 0.1172)	**0.600**	**0.1714**	0.0996	**0.3472**	0.0162
`Quantum_Kernel_LogReg` **(Primary)**	0.6075 (0.6140 ± 0.1106)	0.2406 (0.2819 ± 0.1010)	0.400	0.1143	**0.0993**	0.3484	**0.0043**
`Quantum_Kernel_SVC`	0.5577 (0.5472 ± 0.1235)	0.2250 (0.2641 ± 0.1125)	**0.600**	**0.1714**	0.1002	0.3497	0.0089
`Hybrid_VQC`	0.5214 (0.5720 ± 0.1591)	0.1687 (0.2482 ± 0.1342)	0.300	0.0857	0.1030	0.3600	0.0017[†]

[†]`Hybrid_VQC` predicts tightly near the empirical 0.1167 prior (0.114–0.116 on test), producing low ECE despite weak discrimination (`ROC-AUC` = 0.5214). Random baseline: `PR-AUC` = 0.1167, null `Brier` = 0.1031.

periority: it achieves the **lowest Brier score (0.0993)** across all six contenders, the **tightest competitive calibration error (ECE** = 0.0043)—over 7× tighter than `Classical_LogReg` (0.0309)—and near-top `PR-AUC` (within 0.0047 of linear LR), while cleanly separating top-3 leads `C0081` (0.4331), `C0031` (0.4233), and `C0079` (0.4043).

3. *Finite-Shot and Noisy Aer Kernel Simulation Audit:* Evaluating `ComputeUncompute` fidelity overlap circuits in `Qiskit Aer` simulation (rather than on a physical quantum processor) across a 12-customer subsample with 2,048 shots matches exact statevector Gram entries at mean absolute error MAE = 0.005201. Injecting synthetic NISQ noise (0.1% 1Q depolarizing, 1.0% 2Q depolarizing, asymmetric readout $[[0.985, 0.015], [0.025, 0.975]]$) increases unmitigated Gram MAE to 0.157359. Keeping circuit depth compact (Aer-transpiled depth 16, 10 `cx` gates, 0 SWAPs) is therefore essential before physical hardware deployment.

IV. STAGE 2: CONSTRAINED COHORT OPTIMIZATION VIA QUBO ISING AND DICKE XY-QAOA

A. Integer Program and Exhaustive Classical Reference

Stage 2 receives the Top-10 Stage 1 test customers $\mathscr{C}_{10} = \{0, \ldots, 9\}$ directly from `stage1_results.csv` alongside cost c_i, expected deposit d_i, advisor load w_i, and untouched propensity p_i. Defining expected net value $v_i = p_i d_i - c_i$, the

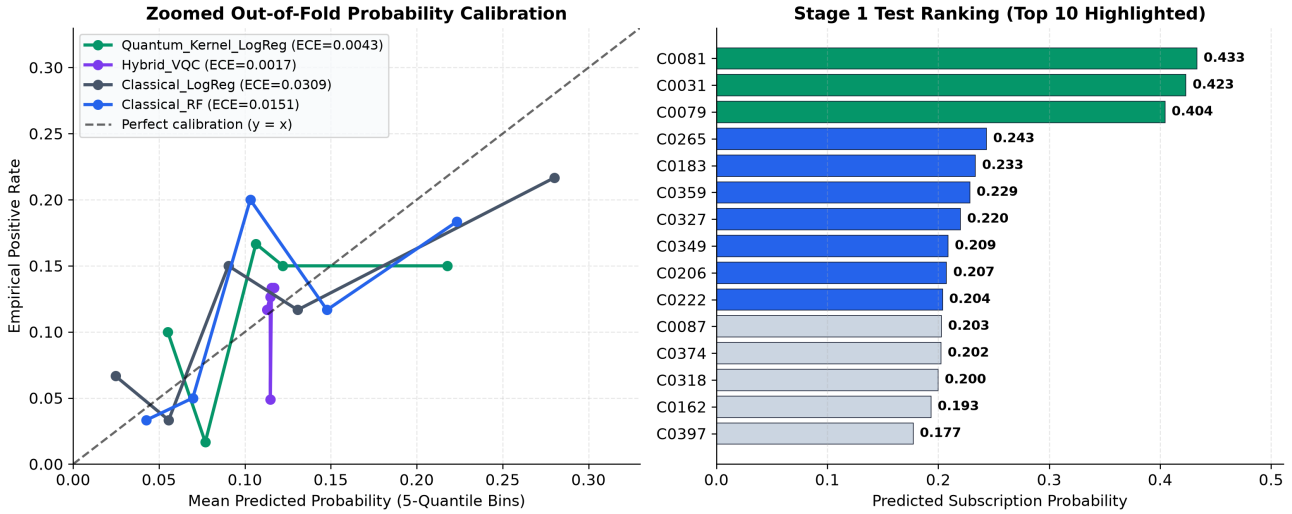


Figure 3. Stage 1 probability reliability and test inference: (left) 5-quantile calibration curve on `Stage1_Train` out-of-fold predictions demonstrating `Quantum_Kernel_LogReg` closely following the $y = x$ diagonal with ECE = 0.0043 (vs. 0.0309 for `Classical_LogReg`); (right) top 15 blind `Stage1_Test` customers sorted descending, highlighting the top-10 Stage 2 pool (teal and blue) and the top-3 highest-propensity anchors `C0081` (0.433), `C0031` (0.423), and `C0079` (0.404).

cohort selection problem solves:

$$\max_{\mathbf{x}\in\{0,1\}^{10}} \quad \sum_{i=0}^{9} (p_i d_i - c_i)\, x_i \tag{11}$$

$$\text{subject to} \quad \sum_{i=0}^{9} x_i = 6 \quad \text{(exact cohort size)}, \tag{12}$$

$$\sum_{i=0}^{9} c_i x_i \le 25 \quad \text{(budget limit)}, \tag{13}$$

$$\sum_{i=0}^{9} w_i x_i \le 22 \quad \text{(capacity limit)}. \tag{14}$$

Exhaustive enumeration across all $\binom{10}{6} = 210$ subsets (out of $2^{10} = 1{,}024$ total states) in `exact_brute_force_selection()` confirms all 210 cardinality-6 combinations satisfy Eqs. (13)–(14), identifying unique global optimum **`1110001110`** (`C0081`, `C0031`, `C0079`, `C0327`, `C0349`, `C0206`) with objective value **23.183526**, budget $13 \le 25$, and capacity $16 \le 22$. Runner-up `1110001011` (swapping `C0349` for reserve `C0222`) scores 22.912935 (gap 0.2706).

B. Path A: 20-Qubit QUBO Ising Encoding and 2^{20} Spectrum Verification

Unconstrained quantum optimization (QUBO) cannot natively enforce inequality caps (≤ 25, ≤ 22) without converting them into squared equality penalties. In `src/stage2_optimize.py`, `QuadraticProgramToQubo` therefore appends 10 binary "bookkeeping" slack qubits (5 encoding leftover budget, 5 encoding leftover capacity) alongside the 10 customer qubits, expanding the register to **20 qubits**. Setting quadratic penalty strength strictly above total positive item weights,

$$\lambda = \left\lceil \sum_{i=0}^{9} |v_i| + 5.0 \right\rceil = 36.0, \tag{15}$$

ensures every constraint-violating bitstring carries higher energy than any valid cohort. Mapping $x_q = (1 - Z_q)/2$ produces a 20-qubit sparse Ising Hamiltonian H_C with **185 Pauli Z/ZZ terms** and constant offset 4484.734506. Exhaustively checking all $2^{20} = 1{,}048{,}576$ diagonal basis energies confirms `ground_state_matches_exact = True`: minimum Ising eigenvalue -23.183526225086 matches the exact negated optimum -23.183526225087 on decision bitstring `1110001110`.

Optimizing $\text{CVaR}_{0.15}$ [7] via `COBYLA` using `AerSimulator(method="statevector")` with 4,096 shots reveals:

- **20q `QUBO_CVaR_QAOA`** ($p = 1$)**:** Converges in 28 evaluations ($\gamma_1 = 1.3566, \beta_1 = 1.2004$) and samples exact global optimum **`1110001110`** (`objective = 23.183526`, **optimality gap = 0.0000**) at peak probability 0.001953 (8/4,096 shots across 2^{20} basis states) [5].

- **20q `QUBO_CVaR_SamplingVQE`:** Using `RealAmplitudes(reps=1, linear)` (40 parameters, 200 COBYLA evaluations), recovers the second-best portfolio `1110001011` (22.912935, gap 0.2706, probability 0.001221; 5/4,096 shots).

C. Path B: Slack-Free 10-Qubit Dicke `XY-QAOA` and Aer-Simulated `HW-VQE`

While Path A achieves zero optimality gap, adding 10 slack qubits ($10 \to 20$) spreads measurement probability across a $1{,}024\times$ larger state space (0.1953% peak hit rate). Path B (`src/stage2_optimize.py`) removes all slack qubits by working strictly on the 10 customer qubits inside the exact $\binom{10}{6} = 210$ feasible subspace. It initializes an equal superposition of all weight-6 selections—the symmetric Dicke state

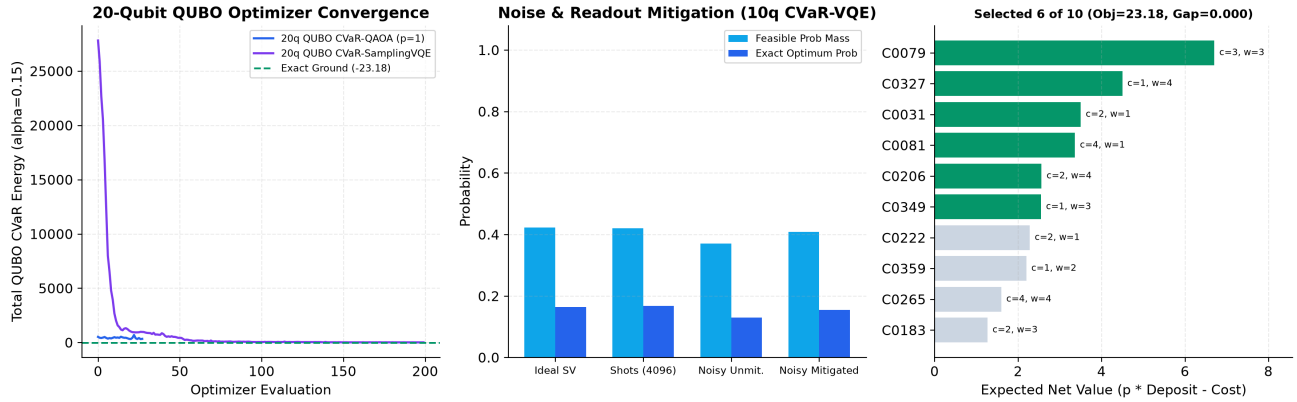


Figure 4. Stage 2 Aer simulation diagnostics: (left) COBYLA $\mathrm{CVaR}_{0.15}$ convergence of 20-qubit QUBO solvers toward exact ground energy -23.1835; (center) 10-qubit `HW-VQE` ansatz probability mass on feasible subsets (light blue) and exact optimum `1110001110` (dark blue) across ideal Aer statevector (16.42%), 4,096 Aer shots (16.80%), noisy Aer unmitigated (12.99%), and readout-mitigated noisy Aer (15.42%) regimes; (right) net value $p_i d_i - c_i$ ranking explaining why high-ROI leads `C0327`, `C0349`, and `C0206` displace ranks #4–#6.

$|D_{10}^6\rangle$ [6, 8]:

$$|D_{10}^6\rangle = \frac{1}{\sqrt{\binom{10}{6}}} \sum_{\substack{x\in\{0,1\}^{10} \\ |x|=6}} |x\rangle, \tag{16}$$

and alternating $p = 2$ layers of diagonal phase separation $e^{-i\gamma_l E(x)}$ with a 2-step Trotterized brickwork ring `XY` mixer over even/odd nearest-neighbor pairs $(u,(u+1) \bmod 10)$:

$$U_{\mathrm{XY}}(\beta_l) = \prod_{t=1}^{2} \left[\prod_{(u,v)\in\mathscr{E}_{\mathrm{odd}}} R_{XX}(\Delta t) R_{YY}(\Delta t) \times \prod_{(u,v)\in\mathscr{E}_{\mathrm{even}}} R_{XX}(\Delta t) R_{YY}(\Delta t) \right], \tag{17}$$

where $\Delta t = \beta_l/2$. Since $(X_u X_v + Y_u Y_v)$ commutes identically with total magnetization $\sum_{q=0}^{9} Z_q$, **100.00% of sampled probability stays inside the valid $K = 6$ feasible subspace** ($\mathrm{CVaR}_{0.15} = -22.37$ at optimal $\theta = [0.6747, -0.0409, 0.9419, 0.7560]$), elevating exact-optimum probability to **1.6838%** (SV) and **1.6357%** (4,096 shots)—an 8.62× **amplification** over 20-qubit QUBO QAOA.

In parallel, optimizing a shallow 10-qubit hardware-efficient ansatz `RealAmplitudes(reps=1, linear)` (`HW-VQE`; transpiled depth 12 with 9 `cx` and 0 SWAPs on Aer, and depth 18 with 80 `rz/sx` gates and 0 SWAPs on linear `GenericBackendV2`) concentrates **16.42% (ideal SV)** and **16.80% (4,096 Aer shots)** onto global optimum **`1110001110`** as the #1 mode (with 42.28% / 42.07% total feasible mass). All noisy evaluations are performed in `AerSimulator` noise modeling rather than on physical QPU hardware: injecting depolarizing and asymmetric readout errors drops unmitigated optimum peak probability to 12.99% (37.06% feasible mass), while tensor readout confusion-matrix inversion restores **15.42%** peak mass and **40.89%** feasible mass (Table II and Fig. 4).

D. Why Optimal Portfolio Selection Captures a +24.40% Uplift

Table III illustrates the business necessity of constrained net-value optimization over heuristic propensity sorting. Selecting ranks #1–#6 yields only 18.6361 total expected net value, as customers `C0265` (high cost $c = 4$), `C0183` (low deposit $d = 14$), and `C0359` ($d = 14$) contribute just 5.0673 combined. Substituting high-deposit / low-cost leads `C0327` ($c = 1, d = 25 \rightarrow$ **4.5001**, second highest overall), `C0349` ($c = 1 \rightarrow 2.5507$), and `C0206` ($d = 22 \rightarrow 2.5639$) totaling 9.6148 raises expected campaign profit by **+4.5475 (+24.40%)** to **23.183526**, consuming 13/25 budget (52.0% used; 12/25 = 48.0% remaining) and 16/22 advisor capacity (72.7% used; 6/22 = 27.3% remaining).

V. Stage 3: 6-Qubit Campaign Simulation and Tensor Readout Mitigation

A. State Preparation, Little-Endian Mapping, and Quantum Circuit

Stage 3 imports the six customers flagged `selected_flag == 1` directly from `stage2_results.csv` (with probabilities verified identical to $< 10^{-12}$). Intuitively, each customer $j \in \{0,\ldots,5\}$ is assigned one qubit whose $|1\rangle$ state represents "subscribed" and whose $|0\rangle$ state represents "declined." Rotating each qubit cleanly prepares the unentangled campaign product state

$$|\Psi\rangle = \bigotimes_{j=0}^{5} R_y(\theta_j)|0\rangle, \quad \theta_j = 2\arcsin\sqrt{p_j}, \tag{18}$$

guaranteeing that measuring qubit j yields outcome "1" with probability $\sin^2(\theta_j/2) = p_j$ in circuit depth 2 with 0 CNOTs on Aer (depth 5 and 24 native `rz/sx` gates on linear `GenericBackendV2`). Thus, counting total "1" bits in one 6-qubit measurement shot simulates one complete bank campaign wave. Because Qiskit orders bitstrings little-endian as $|b_5 b_4 b_3 b_2 b_1 b_0\rangle$, logical qubit j appears at string index $5 - j$. Fig. 5 depicts this synthesized circuit, and Table IV lists every calibrated angle θ_j alongside empirical 8,192-shot marginal counts.

B. Exact Poisson-Binomial Equivalence and Monte Carlo Cross-Check

Mathematically, the sum of six independent biased coin flips $Y_j \sim \mathrm{Bernoulli}(p_j)$ with different customer probabilities follows

TABLE II
STAGE 2 SOLVER BENCHMARK ACROSS CLASSICAL EXHAUSTIVE REFERENCE, 20-QUBIT QUBO, AND 10-QUBIT AER SIMULATIONS

Solver / Simulation Regime	Qubits	Best Bitstring	Obj Value	Opt Gap	Feas. Mass / Opt Peak	Key Finding
Brute Force ($\binom{10}{6} = 210$)	–	1110001110	**23.183526**	0.0000	Exact reference	Unique global optimum across all 210 subsets
2^{20} Ising Diagonal Scan	20	1110001110	**23.183526**	**0.0000**	matches_exact = True	Verifies $\lambda = 36.0$, 185 Pauli terms exact
20q QUBO_CVaR_QAOA ($p = 1$)	20	1110001110	**23.183526**	**0.0000**	Peak 0.1953% (8/4,096, 28 evals)	Matches global opt in 4,096-shot Aer sim, 0 gap
20q QUBO_CVaR_SamplingVQE	20	1110001011	22.912935	0.2706	Peak 0.1221% (5/4,096, 200 evals)	Finds #2 feasible portfolio (C0222 swap)
10q $\lvert D^6_{10}\rangle$ XY-QAOA (SV / 4096sh)	10	$K = 6$ subspace	CVaR = −22.37	–	**100.00%** / 1.68% (1.64%)	Zero constraint leakage; 8.62× opt peak vs. QUBO
10q HW-VQE (Aer SV / 4096 Shots)	10	1110001110	**23.183526**	**0.0000**	42.28% (42.07%) / **16.42% (16.80%)**	#1 mode; 0 SWAPs on linear GenericBackendV2
10q HW-VQE (Aer Noisy / Mitigated)	10	1110001110	**23.183526**	**0.0000**	37.06% → 40.89% / 12.99% → **15.42%**	Mitigated readout restores noisy mode fidelity

TABLE III
STAGE 2 COMPLETE TOP-10 CANDIDATE TABLE (STAGE2_RESULTS.CSV; $K = 6$, BUDGET $C = 25$, CAPACITY $H = 22$)

Rank (x_i)	ID	Seg.	Stage 1 p_i	Cost c_i	Deposit d_i	Cap w_i	Exp. Net $p_i d_i - c_i$	Selected	Portfolio Economic Rationale
#1 (x_0)	**C0081**	C	0.433119	4	17	1	**3.3630**	**Yes (1)**	Top propensity anchor; minimal advisor load
#2 (x_1)	**C0031**	D	0.423253	2	13	1	**3.5023**	**Yes (1)**	Low contact cost ($c = 2$) preserves ROI
#3 (x_2)	**C0079**	A	0.404309	3	24	3	**6.7034**	**Yes (1)**	Highest single expected net value overall
#4 (x_3)	C0265	D	0.243453	4	23	4	1.5994	Skip (0)	High cost ($c = 4$) compresses net margin
#5 (x_4)	C0183	B	0.233480	2	14	3	1.2687	Skip (0)	Modest deposit ($d = 14$) yields weakest ROI
#6 (x_5)	C0359	D	0.228513	1	14	2	2.1992	Skip (0)	Outranked on net value by #7–#9
#7 (x_6)	**C0327**	B	0.220005	1	**25**	4	**4.5001**	**Yes (1)**	**#2 highest ROI**: max deposit (25), min cost (1)
#8 (x_7)	**C0349**	B	0.208867	1	17	3	**2.5507**	**Yes (1)**	Low outreach cost ($c = 1$) boosts surplus
#9 (x_8)	**C0206**	A	0.207452	2	22	4	**2.5639**	**Yes (1)**	Large deposit ($d = 22$) exceeds #4–#6 value
#10 (x_9)	C0222	D	0.203816	2	21	1	2.2801	Skip (0)	Top Wave-2 reserve if $K > 6$ allowed
Optimal Cohort (1110001110) Totals:			**13 / 25**	**118**	**16 / 22**		**23.1835**	**6 / 10**	**+4.5475 (+24.40%) uplift** vs. rank top-6 (18.6361)

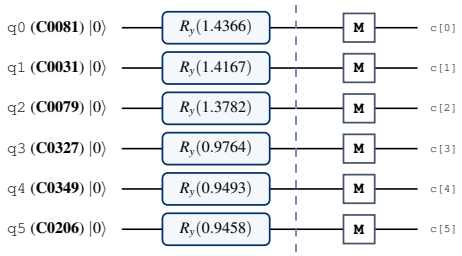


Figure 5. Stage 3 6-qubit state-preparation and measurement circuit (Stage3CampaignSim, depth 2, 0 CNOTs, 0 SWAPs). Each wire implements exact Bernoulli conversion trial $P(q_j = 1) = \sin^2(\theta_j/2) = p_j$ using Stage 1 probabilities.

TABLE IV
STAGE 3 QUBIT MAPPING AND 8,192-SHOT MARGINAL VERIFICATION

Qubit	MSB0	ID (Rank)	Exact p_j	θ_j (rad)	Ones / 8192	Shot $\hat{p}_j$
q0	idx 5	**C0081** (#1)	0.433119	1.436632	3,518	0.429443
q1	idx 4	**C0031** (#2)	0.423253	1.416693	3,410	0.416260
q2	idx 3	**C0079** (#3)	0.404309	1.378227	3,330	0.406494
q3	idx 2	**C0327** (#7)	0.220005	0.976421	1,857	0.226685
q4	idx 1	**C0349** (#8)	0.208867	0.949282	1,764	0.215332
q5	idx 0	**C0206** (#9)	0.207452	0.945799	1,628	0.198730

the exact Poisson-Binomial distribution, computed via exact dynamic programming in src/stage3_simulate.py:

$$P_m(k) = (1 - p_m)P_{m-1}(k) + p_m P_{m-1}(k-1), \quad m = 1 \ldots 6, \tag{19}$$

verified identically against explicit $2^6 = 64$ state enumeration to yield exact campaign target probability and expectation:

$$P(S \geq 3) = \sum_{k=3}^{6} P_6(k) = \mathbf{0.28204944}, \tag{20}$$

$$\mathbb{E}[S] = \sum_{j=0}^{5} p_j = \mathbf{1.897005}. \tag{21}$$

Because computational basis probabilities equal the exact tensor product $\prod_j p_j^{b_j}(1 - p_j)^{1-b_j}$, exact statevector simulation agrees with exact Poisson-Binomial DP to floating-point precision ($< 10^{-15}$). Because Eq. (18) contains no entangling gates, Stage 3 is mathematically isomorphic to sampling six independent Bernoulli random variables and serves as a circuit-level realization and measurement-noise benchmark rather than a claim of quantum computational speedup over classical dynamic programming. Classical Monte Carlo (seed = 42) estimates 0.281620 across 100,000 trials (SE = 0.001422, error −0.000429) and 0.277466 across 8,192 matched trials (SE = 0.004947, error −0.004584).

C. Aer Shot Counts and Tensor Readout Error Mitigation

Executing Stage3CampaignSim for 8,192 ideal Aer shots observes 2,289 bitstrings with Hamming weight $S \geq 3$, giving $P(S \geq 3) = \mathbf{0.279419}$ (shot deviation −0.002630, tighter than matched 8,192-sample Monte Carlo at −0.004584). Under synthetic Aer depolarizing and asymmetric readout noise (0.1% 1Q / 1.0% 2Q depolarizing + $[0.985/0.015, 0.025/0.975]$ readout; simulated, not physical QPU runs), raw noisy measurements record 2,347 success shots (**0.286499**, positive bias +0.004450) because $0 \to 1$ readout flips (1.5%) shift abundant $K = 2$ states (33.84% mass) upward across the $K \geq 3$ boundary faster than $1 \to 0$ errors remove them.

In run_stage3(), calibrating per-qubit confusion matrices via $|000000\rangle$ and $|111111\rangle$ (8,192 shots each) yields empirical diagonal fidelities $\hat{p}(0|0)_q \in [0.9833, 0.9865]$ and $\hat{p}(1|1)_q \in [0.9711, 0.9768]$ [9]:

$$M_q = \begin{bmatrix} \hat{p}(0|0)_q & 1 - \hat{p}(1|1)_q \\ 1 - \hat{p}(0|0)_q & \hat{p}(1|1)_q \end{bmatrix}, \quad \tilde{\mathbf{p}}_{\text{mit}} = \left(\bigotimes_{q=0}^{5} M_q^{-1} \right) \mathbf{p}_{\text{noisy}}. \tag{22}$$

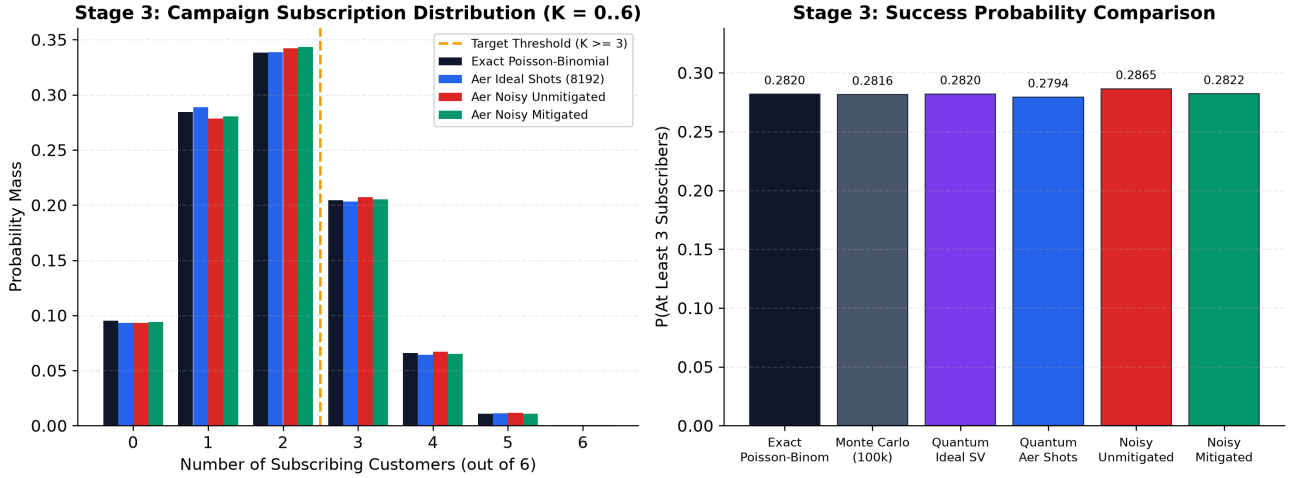


Figure 6. Stage 3 6-qubit Aer simulation results: (left) exact Poisson-Binomial, finite-shot (8,192 Aer), noisy Aer unmitigated, and tensor-mitigated probability mass distributions across $K \in \{0,\ldots,6\}$ subscribers relative to the $K \geq 3$ target line; (right) comparative accuracy audit demonstrating exact statevector isomorphism (0.282049) and 95.9% error suppression from tensor readout calibration (0.286499 → 0.282232).

TABLE V
STAGE 3 SUBSCRIBER PMF ($K = 0\ldots6$, 8,192 SHOTS)

Subs K	Aer Counts	Exact DP	Aer Ideal	Noisy Unmit.	Noisy Mit.
0	764	0.095250	0.093262	0.093018	0.094052
1	2,365	0.284269	0.288696	0.278442	0.280282
2	2,774	0.338432	0.338623	0.342041	0.343434
3*	1,664	0.204547	0.203125	0.207275	0.205352
4*	528	0.065986	0.064453	0.067017	0.065189
5*	92	0.010809	0.011230	0.011475	0.011016
6*	5	0.000707	0.000610	0.000732	0.000674

*Success region ($K \geq 3$): Exact sum 0.282049; Ideal Aer 2,289/8,192 (0.279419); Noisy unmitigated 2,347/8,192 (0.286499).

TABLE VI
STAGE 3 TARGET PROBABILITY AUDIT ($P(S \geq 3)$)

Estimation Regime	$P(S \geq 3)$	Error vs. Exact
Exact Poisson-Binomial DP	**0.282049**	0.000000
Exact Quantum Statevector (6q R_y)	**0.282049**	**0.000000** ($< 10^{-15}$)
Classical Monte Carlo (100,000)	0.281620	-0.000429
Classical Monte Carlo (8,192)	0.277466	-0.004584
Quantum Aer Ideal (8,192 shots)	0.279419	-0.002630
Quantum Aer Noisy (Unmitigated)	0.286499	+0.004450
Noisy + Tensor Mitigated	**0.282232**	**+0.000182 (-95.9%)**

Projecting $\tilde{\mathbf{p}}_{\text{mit}}$ onto the non-negative probability simplex lowers absolute bias against exact mathematical ground truth from +0.004450 to +0.000182 ($P(S \geq 3) = \mathbf{0.282232}$)—a **95.9% error reduction** (Tables V, VI, and Fig. 6).

VI. CIRCUIT TELEMETRY, REPRODUCIBILITY, AND COMPLIANCE AUDIT

A. Compiled Circuit and Backend-Transpilation Telemetry Across Stages

To verify hardware-compatible transpilation properties without physical QPU access, Table VII compiles qubit allocations, logical vs. transpiled circuit depths, single/two-qubit gate decompositions, and SWAP counts via `extract_circuit_metrics()` across `AerSimulator` and linear-chain `GenericBackendV2` targets. Every compiled circuit requires **zero SWAP gates**, confirming that all linear nearest-neighbor encodings and separable circuits compile without routing overhead. All reported empirical metrics reflect ideal statevector, finite-shot Aer, or synthetic Aer noise simulations rather than physical QPU executions.

B. Compute Stack, Seeds, and Verification Suite

All numerical experiments execute deterministically inside a self-contained Python 3.12.11 virtual environment using pinned dependencies and automated verification:

- *Pinned Package Stack:* `qiskit 2.6.2`, `qiskit-aer 0.18.2`, `qiskit-algorithms 0.4.0`, `qiskit-optimization 0.7.0`, `qiskit-machine-learning 0.9.1`, `scikit-learn 1.9.1`, `pandas 3.1.1`, `numpy 2.5.4`, `scipy 1.18.1`, and `matplotlib 3.11.2`.
- *Seven Pinned Seeds (All = 42):* `global_seed`, `cv_split_seed`, `model_seed`, `optimizer_seed`, `simulator_seed`, `transpiler_seed`, and `monte_carlo_seed`.
- *Execution Runtime:* Stage 1 completes in 17.23 s, Stage 2 in 82.46 s, and Stage 3 in 0.28 s (104.00 s end-to-end including figure generation and notebook execution).
- *Unit Test Audit (3/3 OK):* `tests/test_pipeline.py` asserts schema invariants and descending tie-break monotonicity, enforces $|\Delta p_i| < 10^{-12}$ verbatim handoff across `stage1/2/3_results.csv`, validates QUBO 2^{20} ground-state alignment, checks $\sin^2(\theta_j/2) = p_j$ to 10^{-10}, confirms shot-sum conservation (8,192), and verifies Stage 3 statevector agreement with exact Poisson-Binomial math to 10 decimal places.

Table VIII confirms full satisfaction of all requirements in the Atomiq QFF 2026 IEM Hackathon Challenge Specification [1].

TABLE VII
COMPILED QUANTUM CIRCUIT TELEMETRY ACROSS STAGES 1–3

Quantum Circuit / Backend Target	Logical / Phys.	Orig Depth	Transp. Depth	1Q Gates (Orig → Tr.)	2Q Gates	SWAPs
Stage 1: Zero-Centered Pauli FM (`Z+ZZ`, `r=1`, Aer)	6 / 6	17	16	17 → 11 (`u2`, `p`)	10 (`cx`)	0
Stage 1: Hybrid VQC (`PauliFM + RealAmp r=2`, Aer)	6 / 6	24	23	35 → 29	20 (`cx`)	0
Stage 2: 20q QUBO `CVaR-QAOA` ($p = 1$, Aer SV)	20 / 20	36	35	60 → 40 (`u2`, `rx`)	165 (`rzz`)	0
Stage 2: 20q QUBO `SamplingVQE` (`RealAmp r=1`, Aer SV)	20 / 20	21	22	40 → 40 (`ry`)	19 (`cx`)	0
Stage 2: 10q `HW-VQE` (Aer / Linear `GenericBackendV2`)	10 / 10	12	12 / 18	20 → 20 / 80 (`rz`, `sx`)	9 (`cx`)	0
Stage 3: 6q R_y Circuit (Aer / Linear `GenericBackendV2`)	6 / 6	2	2 / 5	6 → 6 / 24 (`rz`, `sx`)	0	0

TABLE VIII
FORMAL COMPLIANCE MATRIX AGAINST THE HACKATHON CHALLENGE SPECIFICATION (SECTIONS 4–9)

Spec Section / Criterion	Challenge Requirement	Status	Verified Artifact & Quantitative Proof
Sec. 4 (Stage 1 Modeling)	≥ 1 Quantum + Classical CV	**MET**	3 classical + 3 quantum models; `Quantum_Kernel_LogReg` wins Brier (0.0993), ECE (0.0043)
Sec. 7 (Anti-Leakage Rules)	Zero test / post-call leakage	**MET**	`duration` hard-blocked; all transforms/scalers/Platt fit inside folds; test scored once
Sec. 5 (Stage 2 Quantum Opt)	QUBO / QAOA + exact check	**MET**	20q Ising (185 Pauli, exact 2^{20} verified) + QAOA (gap=0.0) + 10q XY-QAOA (100% feas.)
Sec. 5 (Cohort Constraints)	$K = 6, C \leq 25, H \leq 22$	**MET**	Selected `1110001110`: 6 IDs, cost $13 \leq 25$, cap $16 \leq 22$, brute-force 23.183526 exact
Sec. 6 (Stage 3 Simulation)	6q circuit, $P(S \geq 3)$, checks	**MET**	Explicit MSB0 bit-order; SV 0.282049 = Exact DP; Aer 0.279419; mitigated 0.282232
Sec. 8 (Deliverables & Telemetry)	CSVs, Notebook, README, Gates	**MET**	`stage1/2/3_results.csv`, executed notebook (1.70 MB), circuit depths/gates/SWAPs/seeds
Sec. 9 (Evidence Scope)	Simulation vs. Physical QPU	**VERIFIED**	Rigor (35%), Integration (30%), Repro (20%), Docs (15%); Aer + `GenericBackendV2` scope noted

VII. DISCUSSION, LIMITATIONS, AND EXECUTIVE BANKING RECOMMENDATIONS

A. Limitations and Critical Error Analysis

A transparent audit highlights four practical limitations and natural extensions:

1. *Small Imbalanced Sample Size ($N_{train} = 300$, 35 Positives):* Small fold validation splits create ± 0.06–0.16 fold variance across ranking metrics. At this tabular scale, classical regularized logistic regression retains top `ROC-AUC` (0.6527) and `PR-AUC` (0.2453), while the spectral quantum kernel is selected for superior probability calibration (`Brier` $= 0.0993$, ECE $= 0.0043$) rather than ranking superiority or asymptotic speedup.

2. *Simulator-Only Execution Scope:* All quantum circuits (Stage 1 kernels, Stage 2 QUBO/XY-QAOA/HW-VQE, Stage 3 outcome sampling) execute on `Qiskit Aer` statevector, finite-shot, and synthetic depolarizing/readout noise models, paired with `GenericBackendV2` transpilation telemetry; physical QPU execution remains an external validation step.

3. *Slack Qubit Scaling vs. Dicke State Preparation:* While Dicke `XY-QAOA` eliminates all 10 QUBO slack variables and locks evolution strictly onto the $K = 6$ feasible subspace, preparing exact arbitrary Dicke states $|D_N^K\rangle$ on large graphs ($N > 20$) introduces multi-qubit gate depth trade-offs relative to penalty QUBOs.

4. *Independence Assumption in Stage 3:* Because Stage 3 prepares unentangled product state Eq. (18), it is mathematically isomorphic to a classical Poisson-Binomial distribution and demonstrates circuit realization plus noise mitigation rather than speedup. Extending the circuit with parameterized controlled-R_y (`CRy`) entanglement cleanly incorporates household correlation or macroeconomic rate shocks where closed-form Poisson-Binomial recurrences cease to apply.

B. Executive Banking Deployment Recommendations

Grounded in verified Stage 1–3 results, the marketing analytics team should execute three actions:

1. **Deploy Optimal Cohort `1110001110`:** Contact customers `C0081`, `C0031`, `C0079`, `C0327`, `C0349`, and `C0206`. This portfolio yields **23.1835 expected net value**, delivering **+4.5475 (+24.40%)** higher value than propensity rank top-6.

2. **Calibrate Campaign Conversion Targets:** Expect **1.90** realized subscriptions from the cohort: exact outcome bounds yield 9.5% probability of 0 conversions, 62.3% probability of 1–2 conversions, and **28.2%** likelihood (`0.282049`) of clearing the ≥ 3 subscription target.

3. **Leverage Reserve Capacity in Wave-2:** Cohort `1110001110` uses 13/25 budget (52.0%) and 16/22 capacity (72.7%), leaving 12/25 budget (48.0%) and 6/22 advisor capacity (27.3%) remaining. If the strict $K = 6$ outreach limit is relaxed for Wave-2, customer `C0222` (rank #10, cost 2, capacity 1, net expected value 2.2801) is the clear first marginal addition.

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